

Path Integration and PoPE: how the two mechanisms compose

Notes derived from the implementation

Abstract

Every positional mechanism here is a choice about one of two factors in the same product. Writing an attention score in polar form as $\mu e^{i\varphi}$, **MapFormer’s path integration determines the positional part of the phase φ** and **PoPE determines the magnitude μ** , which is why the combination (MAPPOPE) is not a new mechanism but the conjunction of two independent choices.

The symmetry is not exact, and the asymmetry matters. MapFormer *adds* a term to the phase and leaves the rest of the attention untouched. PoPE does two things: it replaces the magnitude *and* it eliminates the content-dependent intrinsic phase ψ that ordinary rotary attention carries. Under PoPE each element is a non-negative real times a purely positional phasor, so $\tilde{q}$ can no longer lie anywhere in $\mathbb{C}$. PoPE is therefore a *restriction* on the object MapFormer rotates, not merely a different parameterisation of one factor — and that is a plausible source of the sub-additivity observed on text (§6).

Formulas below are transcribed from the implementation, and where the implementation departs from the published equation or from canonical RoPE that is stated at the point of departure.

1 The shared frame

Let $x_1, \dots, x_T \in \mathbb{R}^d$ be token embeddings, and for a single head write queries and keys $q_t = W_Q x_t$, $k_s = W_K x_s \in \mathbb{R}^{d_{\text{head}}}$. Group the d_{head} coordinates into n_b two-dimensional blocks and identify each block with a complex number,

$$\tilde{q}_t^{(c)} = q_{t,2c} + i q_{t,2c+1}, \quad \tilde{k}_s^{(c)} = k_{s,2c} + i k_{s,2c+1}, \quad c = 0, \dots, n_b - 1. \quad (1)$$

A rotary scheme attaches a per-position angle $\theta_t^{(c)}$ and rotates:

$$\begin{pmatrix} q'_{t,2c} \\ q'_{t,2c+1} \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta_t^{(c)} & -\sin \theta_t^{(c)} \\ \sin \theta_t^{(c)} & \cos \theta_t^{(c)} \end{pmatrix}}_{R(\theta_t^{(c)}) \in SO(2)} \begin{pmatrix} q_{t,2c} \\ q_{t,2c+1} \end{pmatrix}, \quad (2)$$

and likewise for k_s . The unnormalised attention logit is then

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \langle q'_t, k'_s \rangle = \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=0}^{n_b-1} \langle R(\theta_t^{(c)}) q_t^{(c)}, R(\theta_s^{(c)}) k_s^{(c)} \rangle. \quad (3)$$

Because R is orthogonal, $R(\alpha)^\top R(\beta) = R(\beta - \alpha)$, so

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \langle q_t^{(c)}, R(\theta_s^{(c)} - \theta_t^{(c)}) k_s^{(c)} \rangle = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{Re} \left[\overline{\tilde{q}_t^{(c)}} \tilde{k}_s^{(c)} e^{i(\theta_s^{(c)} - \theta_t^{(c)})} \right]. \quad (4)$$

Only the *difference* of angles survives: this is the relative-position property, and it holds for *any* choice of θ .

Finally, write each complex factor in polar form, $\tilde{q}_t^{(c)} = \mu_{t,c}^q e^{i\psi_{t,c}^q}$ and similarly for k . Then (4) becomes

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{\mu_{t,c}^q \mu_{s,c}^k}_{\text{magnitude}} \cos\left(\underbrace{\theta_s^{(c)} - \theta_t^{(c)} + \psi_{s,c}^k - \psi_{t,c}^q}_{\text{phase}}\right). \quad (5)$$

Equation (5) is the frame for everything that follows. **Each mechanism is a decision about where the magnitude comes from and where the phase comes from.**

2 RoPE: the fixed clock

Standard RoPE fixes the angle to be linear in the token index,

$$\theta_t^{(c)} = t \cdot \omega_c, \quad \omega_c = \text{base}^{-2c/d_{\text{head}}} = \text{base}^{-c/n_b}, \quad (6)$$

with ω a non-learnable buffer. The magnitude and the intrinsic phase ψ are whatever W_Q, W_K happen to produce; they are never constrained, and in particular $\tilde{q}$ may lie anywhere in $\mathbb{C}$. Substituting into (4) gives the familiar dependence on $s - t$ alone.

3 MapFormer: path integration replaces the clock

MapFormer keeps (2) and changes only what generates θ .

Step 1 — token to Lie-algebra increment. A low-rank map (rank $r = 2$; paper App. A.7, $W_\Delta = W_\Delta^{\text{out}} W_\Delta^{\text{in}}$) reads *every* token, action or observation:

$$\Delta_t = W_{\text{out}} W_{\text{in}} x_t \in \mathbb{R}^{H \times n_b}, \quad W_{\text{in}} \in \mathbb{R}^{r \times d}, \quad W_{\text{out}} \in \mathbb{R}^{(H n_b) \times r}. \quad (7)$$

Nothing tells the model which tokens are actions; it must *learn* $\Delta \approx 0$ on observations. The paper states only $r \ll d$, “for instance $r = 2$ ” (App. A.7).

Superseded. An earlier note here recorded the bottleneck as load-bearing with $r = 1$ dropping the 2D task to 0.66, and marked it unsupported. It is still unsupported — no $r = 1$ run exists anywhere — but the rank axis is no longer untested: §7 reports an 8-seed sweep over $r \in \{2, 4, 8, 16, 32\}$ in which $r = 2$ is the *worst* setting. Read $r = 2$ below as the paper’s prescription, not as a verified optimum.

Step 2 — integrate.

$$\theta_t^{(h,c)} = \omega^{(h,c)} \sum_{u \leq t} \Delta_u^{(h,c)} = \omega^{(h,c)} \text{cumsum}(\Delta)_t^{(h,c)}. \quad (8)$$

Because $SO(2)$ is abelian, the prefix product of rotations collapses to a *cumulative sum of angles*, which is what makes this an $O(\log T)$ parallel scan rather than a sequential recurrence:

$$\prod_{u \leq t} R(\omega \Delta_u) = R\left(\omega \sum_{u \leq t} \Delta_u\right). \quad (9)$$

Commutativity is the entire computational argument, and it is also the limitation: a turn/forward action space, where displacement depends on accumulated heading, is *not* representable by (8).

Step 3 — frequency schedule. The ω are learnable, initialised geometrically over n_b blocks:

$$\omega_i = \omega_{\max} \left(\frac{1}{\Delta_{\max}} \right)^{\frac{i}{n_b-1}}, \quad i = 0, \dots, n_b - 1, \quad \omega_{\max} = 2\pi, \quad \Delta_{\max} = \text{grid size}. \quad (10)$$

Correction to the published equation. Eq. (10) differs from the paper (v1/v3 eq. 17, v4 eq. 18) in *two* places, both deliberate. First the **sign**: the paper prints a negative exponent, giving an *increasing* schedule that contradicts its own stated boundary condition $\omega_{\min} = \omega_{\max}/\Delta_{\max} \leq \omega_i \leq \omega_{\max}$. Second the **denominator**: the paper writes n_b with $1 \leq i \leq n_b$, so the endpoint $\omega_{\min}$ is never attained; eq. (10) uses $n_b - 1$ with $0 \leq i \leq n_b - 1$, which hits both endpoints exactly. The corrected form is what satisfies the paper’s own inequality and what matches RoPE’s $\theta_i = \text{base}^{-2i/d}$. Both defects survive into v4.

RoPE is a special case — of the functional form, not the encoding. If the model learns $\Delta_t \equiv 1$ for all t , (8) becomes $\theta_t = (t+1)\omega$ (the cumsum is inclusive; only differences enter (4), so the offset is immaterial). That is RoPE’s *form*, with the token index as the clock. It is *not* RoPE identically, for three reasons: ω here is a trained parameter rather than a frozen buffer; its schedule spans $[2\pi/64, 2\pi] = [0.098, 6.28]$ against RoPE’s $[10^{-4}, 1]$, a ratio running from $6.3\times$ at $c = 0$ to $982\times$ at $c = n_b - 1$; and Δ_t is a rank-2 function of the embedding, so $\Delta \equiv 1$ is a hypothesis about what was learned, not a setting. The accurate statement is that path integration is **RoPE with a learned, content-dependent clock rate — which may be zero or negative — over a learned frequency spectrum.**

In terms of (5): MapFormer sets the **phase** and leaves the magnitude untouched.

4 PoPE: an explicit polar parameterisation

PoPE (Polar Coordinate Positional Embeddings) makes (5) explicit rather than emergent.

Notation change, from here on. §1 defined n_b as the number of *2D blocks*, $n_b = d_{\text{head}}/2$. PoPE indexes one frequency per *element*, so throughout this section and the next $n_b = d_{\text{head}}$. Read $\vartheta_c = \text{base}^{-c/n_b}$ below as $\text{base}^{-c/d_{\text{head}}}$; under §1’s definition it would be wrong by a factor of two in the exponent.

Per element c :

$$\mu_{t,c}^q = \text{softplus}(Q_{t,c}), \quad \mu_{s,c}^k = \text{softplus}(K_{s,c}), \quad (\text{content, strictly positive}) \quad (11)$$

$$\varphi_{t,c}^q = t \vartheta_c, \quad \varphi_{s,c}^k = s \vartheta_c + \delta_c, \quad (\text{position, plus a bias}) \quad (12)$$

with $\vartheta_c = \text{base}^{-c/n_b}$ a **fixed buffer** and δ_c a learnable per-(head, frequency) constant.

δ_c is very nearly inert in practice. It is clamped to $[-2\pi, 0]$ in the *forward pass*, not projected onto that range, so its gradient is identically zero outside the interval — and it is initialised at exactly 0, i.e. *on* the upper boundary. Any step that would push it positive kills its gradient permanently. Measured on trained checkpoints (4 heads $\times$ 64 frequencies $\times$ 4 layers), **91–98% of δ_c entries end at ≥ 0** , clamped to zero with dead gradient; per-layer means run -0.0003 to -0.018 rad. For roughly 95% of frequencies, (13) is effectively $\cos((s-t)\vartheta_c)$. Treat the bias as a nominal degree of freedom, not a working one.

The logit is

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \text{softplus}(Q_{t,c}) \text{softplus}(K_{s,c}) \cos((s-t)\vartheta_c + \delta_c). \quad (13)$$

Implemented via the angle-difference expansion, so no complex arithmetic is needed:

$$\cos_{s,c}^K = \cos \varphi_{s,c} \cos \delta_c - \sin \varphi_{s,c} \sin \delta_c, \quad \sin_{s,c}^K = \sin \varphi_{s,c} \cos \delta_c + \cos \varphi_{s,c} \sin \delta_c, \quad (14)$$

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \left[(\mu^q \cos \varphi)_t \cdot (\mu^k \cos^K)_s^\top + (\mu^q \sin \varphi)_t \cdot (\mu^k \sin^K)_s^\top \right]. \quad (15)$$

What PoPE does *not* do. Its phase is a function of position only. The magnitude is content-dependent; the *angle* is not, apart from the constant bias δ_c . This is worth stating because the name invites the opposite reading: PoPE is **not** in the CoPE / Selective-RoPE family, which learn content-dependent *angles*. It modifies the other factor.

In terms of (5): PoPE sets the **magnitude** and leaves the phase as a fixed clock.

5 MapPoPE: the conjunction

Since MapFormer determines φ and PoPE determines μ , combining them requires no new mechanism — only substituting one into the other. Replace PoPE’s positional phase $t\vartheta_c$ with the path-integrated angle (8):

$$a_{ts} = \frac{1}{\sqrt{d_{\text{head}}}} \sum_c \underbrace{\text{softplus}(Q_{t,c}) \text{softplus}(K_{s,c})}_{\text{PoPE: magnitude from content}} \cos \left(\underbrace{\theta_s^{(c)} - \theta_t^{(c)}}_{\text{MapFormer: phase from the path integral}} + \underbrace{\delta_c}_{\text{PoPE bias}} \right), \quad (16)$$

with $\theta_t^{(c)} = \omega^{(c)} \sum_{u \leq t} \Delta_u^{(c)}$ as in (8). In code the class is three lines: inherit MapFormer’s forward pass (so $\cos \theta, \sin \theta$ still come from the path integrator), widen the angle machinery, and swap the attention module.

The one non-trivial detail: frequency count. RoPE rotates *pairs* of coordinates, so it needs $n_b = d_{\text{head}}/2$ angles. PoPE assigns one frequency per *element*, so it needs $n_b = d_{\text{head}}$. The combination therefore rebuilds both position modules at the wider count:

$$n_b : d_{\text{head}}/2 \mapsto d_{\text{head}}, \quad W_{\text{out}} \in \mathbb{R}^{(H d_{\text{head}}) \times r}, \quad \omega \in \mathbb{R}^{H \times d_{\text{head}}}. \quad (17)$$

In parameters this is small: at $d_{\text{model}} = 128$, $H = 2$, one layer, the two models are **204,373** and **204,693**, a difference of +320 (+0.16%) — +64 on ω , +128 on W_{out} , +128 for δ_c . The consequential change is not the parameter count but the **frequency count**: MAPPOPE carries twice as many independent position frequencies, indexed per element rather than per pair. So the two are neither exactly parameter-matched nor frequency-matched, and the second is what a comparison has to contend with.

What this does and does not contaminate. In the $2 \times 2 \{ \text{RoPE}, \text{PoPE} \} \times \{ \text{index}, \text{path-integrated} \}$, both PoPE cells are wide and both RoPE cells narrow, so the width confound is *constant along the encoding factor*. The position main effect and the interaction are therefore

clean; only the encoding main effect carries it — and that is the small one (+0.011 on the torus against +0.461 for position). There is also a direct empirical check: if the extra frequencies were doing the work, PoPE with an index phase should beat RoPE with an index phase. It does not (0.509 vs 0.530, both on the 0.506 floor). The missing control is a *narrow* PoPE at $n_b = d_{\text{head}}/2$ with each angle repeated across its pair, which would isolate the parameterisation from the width; it has not been run.

The widening is a property of PoPE itself, not of the combination — the index-phase PoPE baseline also runs at $n_b = d_{\text{head}}$.

6 Why the two compose, and when they do not

(16) predicts the observed pattern.

- **PoPE alone is inert where position carries the signal.** On the torus navigation task, PoPE with an index phase scores 0.509 against a measured floor of 0.506. A better magnitude parameterisation is worth nothing when the phase encodes no spatial information.
- **PoPE plus path integration is the strongest configuration measured.** 0.994 on the same task. Once the phase carries genuine position, sharpening the content channel pays.
- **On text the picture does not survive its own metric.** Reading the *final* checkpoint gives PoPE alone -0.0053 bpc, path integration alone -0.0041 , sum -0.0093 , both together -0.0058 — apparently sub-additive. But the within-run checkpoint standard deviation is 0.003–0.007, larger than every one of those quantities, which is why the source file specifies the **mean of the last five checkpoints** instead. On that metric the numbers are -0.0041 , -0.0006 , sum -0.0047 , both -0.0052 : path integration alone nearly vanishes and the combination is mildly *super*-additive — the opposite sign. Both single-axis arms are also $n = 1$. **Nothing about additivity on text is established here in either direction**; the two readings are reported so the fragility is visible rather than resolved by choosing one.

The general statement: the two mechanisms are independent as *parameterisations* — they touch disjoint factors of (5) — but that does not make their *effects* independent. Whether they add depends on whether the task has separate work for a phase and a magnitude to do.

7 Selective RoPE, and the family of phase generators

Selective RoPE (Movahedi et al., arXiv:2511.17388, ICLR 2026) was posted on 21 November 2025, three days before MapFormer, and neither paper cites the other in any version. Its rotation angle is, in the notation of (2),

$$\theta_t^{(h,c)} = \tau \sum_{u \leq t} \left[\underbrace{\sigma(W_g x_u)}_{\text{gate}} \odot \underbrace{(\kappa * W_\omega x)_u}_{\text{smoothed increment}} \right]^{(h,c)}, \quad (18)$$

with κ a causal depthwise kernel of width 4 and τ a learned temperature. Set beside (8), both mechanisms put a content-dependent cumulative sum in the phase. This section states the relation exactly, and it is tighter than a family resemblance.

7.1 The generator family

Call a *phase generator* the triple (m, φ, K) producing

$$s_t = s_{t-1} + \varphi(x_{t-K+1}, \dots, x_t) \in \mathbb{R}^m, \quad \theta_t = A s_t \in \mathbb{R}^{Hn_b}. \quad (19)$$

Here m is the **state rank**, φ the increment map, K its temporal support and A a fixed linear readout. Cumulative summation is linear, so it commutes with any linear map and the accumulation may always be pushed down to the smallest space the increment factors through. Three mechanisms are instances:

generator	m	φ	K	A
RoPE	1	$\equiv 1$	1	ω (frozen)
MapFormer	r	$W_{\text{in}}x$	1	$\text{diag}(\omega) W_{\text{out}}$
Selective RoPE	Hn_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	4	τI

For MapFormer the factorisation is an identity, not a reading. Because $\Delta_t = W_{\text{out}}W_{\text{in}}x_t$ and cumsum is linear,

$$\theta_t = \omega \odot \text{cumsum}(W_{\text{out}}W_{\text{in}}x)_t = \omega \odot W_{\text{out}} \underbrace{\sum_{u \leq t} W_{\text{in}}x_u}_{s_t \in \mathbb{R}^r}, \quad (20)$$

checked against the implementation to 9×10^{-7} . **All $Hn_b = 64$ phase channels of MapFormer are linear readouts of a single *two-dimensional* accumulator.** Selective RoPE admits no such factorisation: its 64 channels integrate 64 independent signals.

7.2 The inclusion

The three generators are nested, and each inclusion *removes a constraint* rather than adding a mechanism:

$$\underbrace{\text{RoPE}}_{m=1, \varphi \equiv \text{const}} \subset \underbrace{\text{MapFormer}}_{m=r, \varphi \text{ linear}, K=1} \subset \underbrace{\text{Selective RoPE}}_{m=Hn_b, \varphi \text{ gated}, K=4}.$$

The right-hand inclusion is exact up to a saturating limit: a rank- r W_ω lies in the span of a full-rank one, the delta kernel $\kappa = (0, 0, 0, 1)$ recovers $K = 1$, per-channel scales absorb ω into the rows of W_ω , and $\sigma(\cdot) \rightarrow \text{const}$ as the gate's bias grows. The left-hand inclusion is of functional form only, with the three qualifications recorded in §3.

Consequently **MapFormer's generator is a constrained submodel of Selective RoPE's**, and every comparison between the two is a question about whether a constraint is worth keeping. It is never a question about expressivity, and no experiment here can be read as one.

7.3 Coherence rank

Define the **coherence rank** $\rho = \text{rank}(A) \leq m$: the dimension of the subspace of $\mathbb{R}^{Hn_b}$ that the phase vector is confined to as t varies. It is 1 for RoPE, r for MapFormer, Hn_b for Selective RoPE.

Why this is the load-bearing quantity. Attention observes only $\cos(\theta_s - \theta_t + \psi)$, and cosine is 2π -periodic, so no single block can separate displacements differing by $2\pi/\omega_c$. Position survives only because the blocks are read *jointly* — a residue decoding across the frequency spectrum, whose

ambiguity shrinks with the least common multiple of the periods. That decoding is well posed exactly when the blocks measure the same underlying quantity, i.e. when ρ is small. At $\rho = Hn_b$ there is nothing to decode jointly: one has 64 individually ambiguous single-block codes whose conjunction says nothing about position that each does not say alone.

Two consequences follow, one an impossibility and one an optimisation claim.

(i) Expressivity requires $\rho \geq D$. On a D -dimensional lattice with actions $\pm e_i$, φ is linear, so the state after a trajectory with action counts n_a is $s = \sum_a n_a v_a$ with $v_a = W_{\text{in}} x_a \in \mathbb{R}^r$. Two positions with equal s have identical phase in *every* block. The induced map $\mathbb{Z}^D \rightarrow \mathbb{R}^r$ therefore has a kernel of dimension $D - r$ whenever $r < D$: **$r < D$ makes the position code non-injective by construction** — not badly conditioned but unrepresentable, and no amount of training or depth repairs it.

(ii) $\rho = D$ is expressible but not trainable. In 2D, $r = 2$ is exactly sufficient by (i), and is what the paper prescribes (r is the dimensionality of the action space, App. A.7). It nonetheless costs 0.085 accuracy at $T = 1024$ against $r = 4$ (8/8 seeds, $t = 3.57$), and the code it learns is skewed: opposite actions fail to cancel by half the action scale (0.495) and the two axes are close to parallel ($|\cos| = 0.783$), against 0.092 and 0.175 at $r = 4$. Yet the $r = 4$ latents still place 100.0% of their spectral energy in a 2-plane. **The extra rank carries no signal; it is optimisation slack for realising an exact group homomorphism under gradient descent.** The paper’s structural argument for $r = 2$ is right about what is expressible and wrong about what is reachable.

7.4 What the constraints are worth, measured

One batch, 8 arms, the generator swapped and nothing else changed. Increments are against MapFormer at $r = 2$; MDE is $2.8 \text{sd}/\sqrt{n}$.

constraint removed	params	parity $L=128$	torus $T=1024$
$K = 1 \rightarrow 4$ (causal conv)	+193	−0.020 detectable	−0.064 ($t - 1.99$)
$m = r \rightarrow Hn_b$ (full rank)	+7,873	−0.020 detectable	+0.058 (7/8)
φ linear $\rightarrow$ gated	+8,193	−0.030 detectable	+0.086 (8/8, sign $p=0.008$)
all three (Selective RoPE)	+16,385	−0.009 unmeasured	+0.048 ($t 1.36$)
$r = 2 \rightarrow 4$, <i>constraint kept</i>	+384	—	+0.085 (8/8, $t 3.57$)

Three readings, in decreasing order of how well they are supported.

Rank is the only axis with a measurable effect, and it saturates at 4. $r = 4, 8, 16, 32$ are flat within noise (+0.085 to +0.095), so this is a step at $r = 2$, not a capacity slope. The same gain is available inside MapFormer’s parameterisation for $21\times$ fewer parameters than outside it.

The gate is not doing what its name suggests. It is statistically indistinguishable from the unrelated full-rank addition of comparable size (+0.018 at $T=512$ against an MDE of 0.026), so the parameters and not the mechanism are the plausible cause. The natural mechanism — that a sigmoid suppresses observation tokens, which MapFormer must instead learn to send to $\Delta \approx 0$ — was tested directly and falsified: the gate is $1.35\times$ larger on actions than observations on the torus *where it helps*, and $1.54\times$ larger on the analogous split on parity *where it hurts*.

The convolution is the one knob with a principled sign, and it is negative on both tasks at negligible cost. With $K > 1$ the increment of a token depends on its neighbours, so the phase ceases to be a function of the multiset of tokens traversed and the map from action sequences to $SO(2)^{n_b}$ is no longer a group homomorphism. Commutativity is MapFormer’s entire

computational argument — and the word does not appear once in Selective RoPE. The measured cost of giving it up is -0.020 on parity and -0.064 on the torus for 193 parameters.

7.5 A prediction, and the paper’s own 5D negative

MapFormer v4 Table 6 reports MapWM collapsing in 5D: 0.75/0.50/0.35 on IID/OOD-d/OOD-s, *below* CoPE at 0.94/0.80/0.69, while MapEM-s holds at 1.00/1.00/0.87. The paper reports this without an account of it.

Consequence (i) supplies one. If $r = 2$ was held fixed at $D = 5$ — the paper specifies only $r \ll d$, “for instance $r = 2$ ” — then the position code has a three-dimensional kernel and **the collapse is structurally forced**, a property of the chosen rank rather than of path integration. MapEM’s survival is consistent on the same reading: its multiplicative $A_X \odot A_P$ lets the content channel disambiguate what a degenerate position channel cannot.

Pre-registered test. A D -dimensional torus, $r \in \{2, D, D+2\}$ at $D \in \{3, 5\}$, one batch, 8 seeds.

- $r = 2$ collapses at $D = 5$ and degrades only mildly at $D = 3$ — kernel dimensions 3 and 1.
- $r = D$ recovers most of the loss, being exactly expressible, but retains the $r=2$ conditioning skew (opposition and $|\cos|$ well above their $r=4$ values in 2D).
- $r = D+2$ is clean, and larger r is flat.

Falsified if $r = 2$ is unimpaired at $D = 5$, or if $r = D$ is already flat — either outcome kills the two-threshold reading, and the first would show the 5D collapse has nothing to do with rank.

7.6 The difference that has not been tested

Everything above concerns the *generator*. Selective RoPE also differs in *placement*: it derives the angle per head and per layer from that layer’s own queries, whereas MapFormer computes θ once from the token embeddings and shares it across all layers and heads. The arms here swap the generator only, and at one layer the distinction is nearly vacuous, since $q = W_Q \text{LN}(x)$ is itself a learned linear map of the token. At depth it is real, and it is the sole axis on which the two designs remain uncomparred.

It is also not the axis probed by the refinement experiment of Appendix A: that carried a *bounded gated correction* to θ across iterations of a shared block and the optimiser declined to use it. Recomputing θ from scratch at each layer, from that layer’s representation, is a different model and an open question.

A The other mechanisms, in the same notation

Weight-shared recursion (the “loop”). With B_θ a single transformer block and n the loop count,

$$h^{(0)} = \text{Emb}(x), \quad h^{(j+1)} = B_\theta(h^{(j)}; \cos \theta, \sin \theta), \quad j = 0, \dots, n-1. \quad (21)$$

θ is computed *once* from the token embeddings and reused, matching both parents. Parameter count equals one layer; compute equals n layers.

Carrying a *bounded, gated correction* to θ across iterations — $\theta \leftarrow \theta_0 + g \cdot \tanh(\text{refine}(h))$ with g initialised to zero — is a different model, and it is null: the learned gate settles at mean $|g| \approx 0.083$ with inconsistent sign across seeds, capping the correction at 0.14 rad against a θ spanning $\sim 2\pi T$. The gate was verified escapable (gradient 1.9×10^{-3} at zero), so the optimiser declined rather than failed to express it. A genuine per-iteration recomputation of $\omega \cdot \text{cumsum}(\Delta(h^{(j)}))$ has *not* been run.

Level 1.5 invariant EKF (withdrawn; recorded for completeness). A correction applied to θ before the rotation:

$$R_t = \exp(\text{clamp}(f_R(x_t), -5, 5)) \quad \text{per-token measurement noise} \quad (22)$$

$$z_t = \pi \tanh(f_z(x_t)) \quad \text{measurement in } [-\pi, \pi] \quad (23)$$

$$\nu_t = \text{atan2}(\sin(z_t - \theta_t), \cos(z_t - \theta_t)) \quad \text{wrapped innovation} \quad (24)$$

$$K_t = \Pi / (\Pi + R_t), \quad \Pi = e^{\log \Pi} \quad \text{gain; } \Pi \in \mathbb{R}^{H \times n_b}, \text{ constant in } t \quad (25)$$

$$d_t = (1 - K_t) d_{t-1} + K_t \nu_t, \quad d_{-1} = 0 \quad \text{affine scan} \quad (26)$$

$$\hat{\theta}_t = \theta_t + d_t. \quad (27)$$

Π is a learned ($H \times n_b$) constant — 64 values at the default config, one per head and frequency block — not a scalar; it is constant across *tokens*, which is what distinguishes Level 1.5 from the full heteroscedastic filter. The recurrence is scalar in each channel and affine, so it is one more parallel scan.

Initialisation, which turned out to matter. $\log R$ ’s head is initialised with weights scaled by 0.01 and bias 0, giving $R \approx 1$ and hence $K \approx 0.5$ *at initialisation* — corrections at half strength before anything is learned. On the multiplicative-attention backbone this destroyed the gradient path and diverged 3 of 9 seeds; the fix was a bias of 3.0, i.e. $K \approx 0.05$, making the filter a near-no-op at init.

Status. No individual component is load-bearing: the measurement head, the per-token gate and the learned Π can each be removed alone at no measurable cost ($n = 5$). The whole filter does have an effect, but only at out-of-distribution length (+0.062 at $T=512$, +0.124 at $T=1024$, loss-matched), and on a task built to supply the drift it is meant to correct the effect *does not grow with the drift* — flat in one recipe, negative in another. The mechanism is withdrawn as inference; what survives is stabilisation.

Hierarchy (hourglass). With pooling factor k , a causal *shift-then-mean* pool P_k and an upsample U_k ,

$$h \leftarrow B_{\text{pre}}(h), \quad h \leftarrow h + U_k B_{\text{coarse}}(P_k h), \quad h \leftarrow B_{\text{post}}(h), \quad (28)$$

with the coarse angle pooled alongside the content, and the sequence padded to a multiple of k then truncated after B_{post} .

Two details the notation would otherwise hide. P_k right-shifts by $k - 1$ with a zero pad *before* averaging, so coarse token j is the mean over original indices $jk - 1, \dots, jk + k - 2$ — not the plain group mean, and the first coarse token is dragged toward zero by the pad. And U_k is `repeat_interleave`, which is *not* the adjoint of P_k (that would carry a $1/k$ and the matching shift); it is written U_k rather than $P_k^\dagger$ for that reason.

Setting all three blocks to the *same* shared module gives the weight-shared hourglass: one block’s parameters (204,373, identical to the flat 1-layer model, against 600,917 unshared) with the coarse block’s token reduction retained. Note that every hourglass class ignores `n_layers` and is always this 3-block scaffold, so a flat control must be built at 3 blocks, not 1.

B Empirical anchors

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
PoPE	magnitude μ	$d_{\text{head}}/2 \rightarrow d_{\text{head}}$ widening, +320	+0.027 <i>with</i> path int. Without it, both arms are on the floor — see note
Bottleneck rank	state rank ρ	+384 on 204,373 (+0.19%)	+0.085 torus $T=1024$ for $r=2 \rightarrow 4$ (8/8); flat to $r=32$
Recursion	block reuse	0	+0.056 parity (16/16); matches or beats $3\times$ params
Hierarchy	token count	0 if weight-shared	length-dependent: ≈ 0 at $L = 16$, +0.060 at $L = 512$ (23/24 seeds); -22.8% time at $L = 2048$, but see note (iv)
InEKF	$\theta \rightarrow \hat{\theta}$	+49,600 on 204,373 (+24%)	no individual <i>component</i> is load-bearing; the whole filter is +0.06 to +0.13, at OOD length only

Reading the table. Parity figures are at $L = 128$ trained at $L = 16$, from one batch of $8 \text{ arms} \times 16$ seeds. Torus figures are the position-axis factor average over both encodings (+0.4379 RoPE, +0.4851 PoPE). **Three qualifications the columns cannot carry.** (i) The hierarchy row is *not* a parity-at- $L = 128$ figure: its accuracy comes from runs trained at $L = 16$ and $L = 512$, and its time saving is measured at $L = 2048$ — at $L = 128$ the same benchmark shows +0.4%, i.e. no saving at all. (ii) The InEKF’s parameter cost is measured on a 204,373-parameter torus model and the path-integration cost on a 199,042-parameter parity model; the two percentages are not commensurable, and the InEKF has never been run on parity. (iii) PoPE without path integration is a difference between two models sitting on the measured floor of 0.506 (0.509 vs 0.530), with a seed spread wider than the gap — it is a floor artifact, not an effect, and is omitted from the column for that reason. (iv) The -22.8% is measured on LOOPEDHOURGLASS against the unshared flat scaffold, so it *bundles* the shared block’s parameter saving with the pooling’s compute saving. Isolated on text, hierarchy alone is $1.23\times$ throughput ($\approx -18.7\%$) at -14.1% memory. Attributing -22.8% to the hierarchy axis alone overstates it.